Assessment status: Preliminary. Reviewed and approved as Preliminary.

DEEP Detailed Assessment

Western Gulf Coastal Plain reference assessment

Mink Brook

Scenario: Culvert removed

The road crossing replaced by a span.

Source	USEPA NRSA 2013-24 (L3 ecoregion 34), StreamCurves regional analysis, methodology 0.16
Version / status	v1 · Preliminary
Predictor source	streamcat
Region match	Vermont (50) · 58 Northeastern Highlands
Coordinates	43.68581234, -72.23671234
COMID / HUC8	9327042 / 01080104
Drainage area	12.43 km2
HR reach watershed area	11.98 km2
Reach length	1000.0 ft
Watershed basis	HR reach watershed (STAF site engine v0.5.0)
Desktop values	STAF site engine (HR reach watershed) 4, StreamCat lookup engine (NHDPlus V2 basin) 3
StreamCat reach	COMID 9327042 (this reach)
Content digest	sha256:5e5fa2b7d4658d1a0a9fec1ea953ce9f61c3610fc882279f3a9d2c5354499b79
Ecosystem Condition Index	0.34 (Non-Functioning)
STAF function coverage	20 of 20
Flagged transfers	11 of the 20 rated functions rest on a reference pool whose transfer to this ecoregion was not confirmed by the recovery test (Surface water storage, Streamflow regime, Low flow and baseflow dynamics, Floodplain connectivity, Hyporheic connectivity, Channel evolution, Sediment continuity, Bed composition and bedform dynamics, Light & thermal regime, Carbon processing, Habitat provision); each is rated with confidence capped and flagged on its metric row.

Watershed

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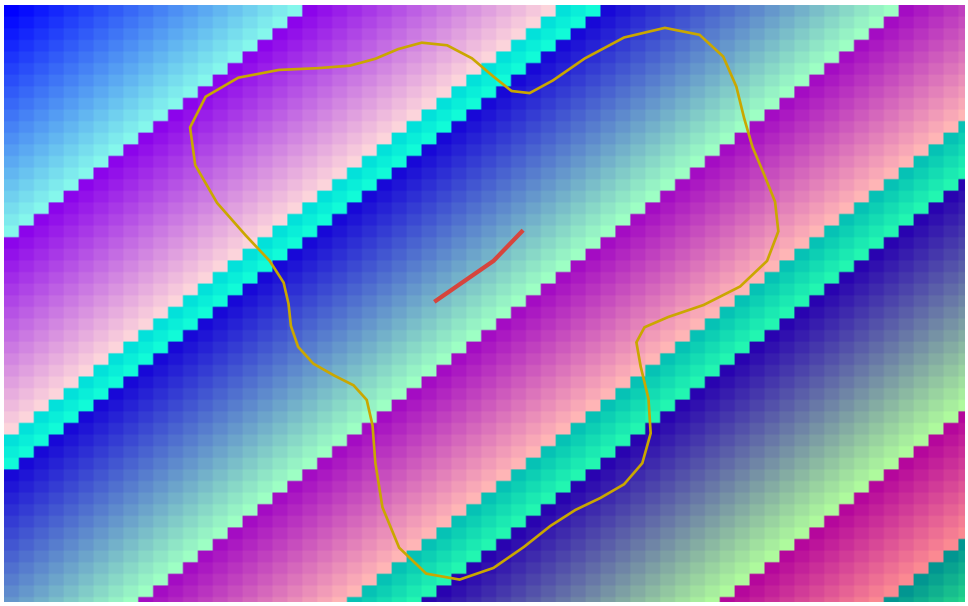

Function scores (0-15)

Function	Score	Condition
Catchment hydrology	13.3	F
Surface water storage		Not scored
Reach inflow	5.9	AR
Streamflow regime	9.2	AR
Low flow and baseflow dynamics	4.5	NF
High flow dynamics	13.6	F
Floodplain connectivity		Not scored
Hyporheic connectivity		Not scored
Channel evolution		Not scored
Channel and floodplain dynamics	12.9	F
Sediment continuity		Not scored
Bed composition and bedform dynamics		Not scored
Light & thermal regime		Not scored
Carbon processing		Not scored
Nutrient cycling	0.0	NF
Water & soil quality		Not scored
Habitat provision		Not scored
Population support		Not scored
Community dynamics	0.1	NF
Watershed connectivity	0.0	NF

Function limitations

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Surface water storage: This function is rated on a curve whose reference pool transfer to this ecoregion was not confirmed by the recovery test (Wetland); the score enters the index with confidence capped.

Streamflow regime: This function is rated on a curve whose reference pool transfer to this ecoregion was not confirmed by the recovery test (Base flow index); the score enters the index with confidence capped.

Low flow and baseflow dynamics: This function is rated on curves whose reference pool transfer to this ecoregion was not confirmed by the recovery test (Base flow index, Residual pool depth); the score enters the index with confidence capped.

Floodplain connectivity: This function is rated on curves whose reference pool transfer to this ecoregion was not confirmed by the recovery test (Bankfull height above channel, Bankfull width:depth ratio); the score enters the index with confidence capped.

Hyporheic connectivity: This function is rated on a curve whose reference pool transfer to this ecoregion was not confirmed by the recovery test (Embeddedness); the score enters the index with confidence capped.

Channel evolution: This function is rated on curves whose reference pool transfer to this ecoregion was not confirmed by the recovery test (Bankfull height above channel, Bankfull width:depth ratio); the score enters the index with confidence capped.

Sediment continuity: This function is rated on a curve whose reference pool transfer to this ecoregion was not confirmed by the recovery test (Relative bed stability); the score enters the index with confidence capped.

Bed composition and bedform dynamics: This function is rated on a curve whose reference pool transfer to this ecoregion was not confirmed by the recovery test (Embeddedness); the score enters the index with confidence capped.

Light & thermal regime: This function is rated on a curve whose reference pool transfer to this ecoregion was not confirmed by the recovery test (Canopy density, mid-channel); the score enters the index with confidence capped.

Carbon processing: This function is rated on a curve whose reference pool transfer to this ecoregion was not confirmed by the recovery test (Riparian veg cover, woody); the score enters the index with confidence capped.

Habitat provision: This function is rated on a curve whose reference pool transfer to this ecoregion was not confirmed by the recovery test (Natural fish cover); the score enters the index with confidence capped.

Outcome sub-indices

Physical	Chemical	Biological	ECI
0.63	0.23	0.15	0.34

Metric measurements & curve indices

Function	Metric	Value	Index	Source	Curve (source)
Catchment hydrology	Impervious surface	3.0	0.91	desktop: STAF site engine v0.5.0 impervious (HR reach watershed, NLCD 2021)	EASI fixed criteria, Catchment hydrology (land-cover pressure)
Catchment hydrology	Agriculture, crop and hay	12.75	0.87	desktop: StreamCat PCTIMP2019WS	EASI fixed criteria, Catchment hydrology (land-cover pressure)
Surface water storage	Wetland	0.4	ref. only	desktop: Computed (reference only)	USEPA NRSA 2013-24 (L3 ecoregion 34), StreamCurves regional analysis, methodology 0.16-provisional
Reach inflow	Road density	3.0	0.39	desktop: STAF site engine v0.5.0 impervious (HR reach watershed, NLCD 2021)	EASI fixed criteria, Road-density proxy for concentrated inflow pressure
Streamflow regime	Base flow index	12.75	0.29	desktop: StreamCat PCTIMP2019WS	USEPA NRSA 2013-24 (L3 ecoregion 34), StreamCurves regional analysis, methodology 0.16-provisional
Streamflow regime	Degree of regulation	0.4	0.94	desktop: Computed	EASI fixed criteria, Flow alteration (degree of regulation)
Low flow and baseflow dynamics	Base flow index	12.75	0.29	desktop: StreamCat PCTIMP2019WS	USEPA NRSA 2013-24 (L3 ecoregion 34), StreamCurves regional analysis, methodology 0.16-provisional

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Function	Metric	Value	Index	Source	Curve (source)
Low flow and baseflow dynamics	Residual pool depth	7.5	0.30	field	USEPA NRSA 2013-24 (L3 ecoregion 34), StreamCurves regional analysis, methodology 0.16-provisional
High flow dynamics	Impervious surface	3.0	0.91	desktop: STAF site engine v0.5.0 impervious (HR reach watershed, NLCD 2021)	EASI fixed criteria, Catchment hydrology (land-cover pressure)
Floodplain connectivity	Bankfull height above channel				USEPA NRSA 2013-24 (L3 ecoregion 34), StreamCurves regional analysis, methodology 0.16-provisional
Floodplain connectivity	Bankfull width:depth ratio				USEPA NRSA 2013-24 (L3 ecoregion 34), StreamCurves regional analysis, methodology 0.16-provisional
Hyporheic connectivity	Embeddedness			field	USEPA NRSA 2013-24 (L3 ecoregion 34), StreamCurves regional analysis, methodology 0.16-provisional
Channel evolution	Bankfull height above channel				USEPA NRSA 2013-24 (L3 ecoregion 34), StreamCurves regional analysis, methodology 0.16-provisional
Channel evolution	Bankfull width:depth ratio				USEPA NRSA 2013-24 (L3 ecoregion 34), StreamCurves regional analysis, methodology 0.16-provisional
Channel and floodplain dynamics	Sinuosity				USEPA NRSA 2013-24 (L3 ecoregion 34), StreamCurves regional analysis, methodology 0.16-provisional
Channel and floodplain dynamics	Bank angle	12.5	0.86	field	USEPA NRSA 2013-24 (L3 ecoregion 34), StreamCurves regional analysis, methodology 0.16-provisional
Sediment continuity	Relative bed stability				USEPA NRSA 2013-24 (L3 ecoregion 34), StreamCurves regional analysis, methodology 0.16-provisional
Bed composition and bedform dynamics	Embeddedness			field	USEPA NRSA 2013-24 (L3 ecoregion 34), StreamCurves regional analysis, methodology 0.16-provisional
Bed composition and bedform dynamics	Large wood volume				USEPA NRSA 2013-24 (L3 ecoregion 34), StreamCurves regional analysis, methodology 0.16-provisional
Light & thermal regime	Canopy density, mid-channel				USEPA NRSA 2013-24 (L3 ecoregion 34), StreamCurves regional analysis, methodology 0.16-provisional
Carbon processing	Riparian veg cover, woody				USEPA NRSA 2013-24 (L3 ecoregion 34), StreamCurves regional analysis, methodology 0.16-provisional

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Function	Metric	Value	Index	Source	Curve (source)
Nutrient cycling	Total nitrogen	17.5	0.00	field	USEPA NRSA 2018-19 regional total nitrogen thresholds, NARS-9 region CPL
Nutrient cycling	Total phosphorus				USEPA NRSA 2018-19 regional total phosphorus thresholds, NARS-9 region CPL
Water & soil quality	Specific conductivity				USEPA NRSA 2018-19 specific conductance benchmarks, NARS-9 region CPL
Habitat provision	Natural fish cover				USEPA NRSA 2013-24 (L3 ecoregion 34), StreamCurves regional analysis, methodology 0.16-provisional
Population support	Fish MMI				USEPA NRSA 2018-19 fish assemblage multimetric index benchmarks, NARS-9 region CPL
Community dynamics	Benthic macroinvertebrate MMI	22.5	0.01	field	USEPA NRSA 2018-19 benthic macroinvertebrate multimetric index benchmarks, NARS-9 region CPL
Watershed connectivity	Mapped dams within one mile	12.75	0.00	desktop: StreamCat PCTIMP2019WS	EASI fixed criteria, Nearby dam-proximity proxy

Scoring advisories

Wetland: engine-computed value shown as reference only: this curve was fitted on StreamCat predictors, so scoring it against an HR reach watershed value would mix training and serving sources

Base flow index: value 12.75 is below the reference pool's range (23.4382 to 71.95); the score follows the seed's sub-reference convention, not reference data

Base flow index: value 12.75 is below the reference pool's range (23.4382 to 71.95); the score follows the seed's sub-reference convention, not reference data

Reference support

Reference condition is set by a fixed landscape-pressure screen, never by a score. 11 borrow comparable least-disturbed stations from a wider region, stating the region and the transfer risk. 2 carry the national reference label: reference curve fitted to comparable least-disturbed stations drawn from the national pool, because this ecoregion and its parents hold too few of their own. 5 carry the published benchmark label: scored against a published criterion rather than against stations from this ecoregion. The criterion carries its own definition of reference, which need not match this assessment's. 10 rest on a flagged transfer: a comparable least-disturbed pool the recovery test did not confirm for this ecoregion, used because no validated source supports the metric here, with transfer risk unvalidated and confidence capped. A metric that no basis supports is not scored, and the function it would have scored is reported as unassessed.

Metric	Scored against	Curve set
Impervious surface	Fixed criteria, the same in every region	
Agriculture, crop and hay	Fixed criteria, the same in every region	
Wetland	Reference curve from 39 least-disturbed stations borrowed from NARS-9 region CPL under the regional screen (watershed agriculture at most 25 percent), 0 of them in this ecoregion. Transfer risk unvalidated (transfer not confirmed by the recovery test)	
Road density	Fixed criteria, the same in every region	
Base flow index	Reference curve from 39 least-disturbed stations borrowed from NARS-9 region CPL under the regional screen (watershed agriculture at most 25 percent), 0 of them in this ecoregion. Transfer risk unvalidated (transfer not confirmed by the recovery test)	
Degree of regulation	Fixed criteria, the same in every region	
Residual pool depth	Reference curve from 20 least-disturbed stations borrowed from NARS-9 region CPL under the regional screen (watershed agriculture at most 25 percent), 0 of them in this ecoregion. Transfer risk unvalidated (transfer not confirmed by the recovery test)	

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Metric	Scored against	Curve set
Bankfull height above channel	Reference curve from 21 least-disturbed stations borrowed from NARS-9 region CPL under the regional screen (watershed agriculture at most 25 percent), 0 of them in this ecoregion. Transfer risk unvalidated (transfer not confirmed by the recovery test)	
Bankfull width:depth ratio	Reference curve fitted to comparable least-disturbed stations drawn from the national pool, because this ecoregion and its parents hold too few of their own.	
Embeddedness	Reference curve from 17 least-disturbed stations borrowed from NARS-9 region CPL under the regional screen (watershed agriculture at most 25 percent), 0 of them in this ecoregion. Transfer risk unvalidated (transfer not confirmed by the recovery test)	
Sinuosity	Reference curve from 17 least-disturbed stations borrowed from NARS-9 region CPL under the regional screen (watershed agriculture at most 25 percent), 0 of them in this ecoregion. Transfer risk high	
Bank angle	Reference curve from 17 least-disturbed stations borrowed from NARS-9 region CPL under the regional screen (watershed agriculture at most 25 percent), 0 of them in this ecoregion. Transfer risk moderate	
Relative bed stability	Reference curve from 20 least-disturbed stations borrowed from NARS-9 region CPL under the regional screen (watershed agriculture at most 25 percent), 0 of them in this ecoregion. Transfer risk unvalidated (transfer not confirmed by the recovery test)	
Large wood volume	Reference curve fitted to comparable least-disturbed stations drawn from the national pool, because this ecoregion and its parents hold too few of their own.	
Canopy density, mid-channel	Reference curve from 56 least-disturbed stations borrowed from Level I ecoregion 9 (Great Plains) under the regional screen (watershed agriculture at most 25 percent), 0 of them in this ecoregion. Transfer risk unvalidated (transfer not confirmed by the recovery test)	All streams (pooled) (default)
Riparian veg cover, woody	Reference curve from 64 least-disturbed stations borrowed from Level I ecoregion 9 (Great Plains) under the regional screen (watershed agriculture at most 25 percent), 0 of them in this ecoregion. Transfer risk unvalidated (transfer not confirmed by the recovery test)	
Total nitrogen	Scored against a published criterion rather than against stations from this ecoregion. The criterion carries its own definition of reference, which need not match this assessment's.	
Total phosphorus	Scored against a published criterion rather than against stations from this ecoregion. The criterion carries its own definition of reference, which need not match this assessment's.	
Specific conductivity	Scored against a published criterion rather than against stations from this ecoregion. The criterion carries its own definition of reference, which need not match this assessment's.	
Natural fish cover	Reference curve from 64 least-disturbed stations borrowed from Level I ecoregion 9 (Great Plains) under the regional screen (watershed agriculture at most 25 percent), 0 of them in this ecoregion. Transfer risk unvalidated (transfer not confirmed by the recovery test)	All streams (pooled) (default)
Fish MMI	Scored against a published criterion rather than against stations from this ecoregion. The criterion carries its own definition of reference, which need not match this assessment's.	
Benthic macroinvertebrate MMI	Scored against a published criterion rather than against stations from this ecoregion. The criterion carries its own definition of reference, which need not match this assessment's.	
Mapped dams within one mile	Fixed criteria, the same in every region	

Metrics not scored

These metrics are not scored. A metric withheld for insufficient reference support has no curve. No pool of least-disturbed stations in this ecoregion or its parents supports it, no national donor pool was shown to transfer, no modeled expectation passed validation, and no published criterion applies.

Metric	Functions	Reason
EPT taxa richness	Community dynamics	Insufficient reference support
Shannon diversity (H')	Community dynamics	Insufficient reference support
Tolerant individuals (%) ~ HBI	Community dynamics	Insufficient reference support
Benthic total taxa richness	Population support	Insufficient reference support
Chlorophyll a	Light and thermal regime	Insufficient reference support

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Metric	Functions	Reason
pH	Water and soil quality	Insufficient reference support
Turbidity	Water and soil quality	Insufficient reference support
Native fish taxa richness	Population support	Insufficient reference support
Substrate size	Bed composition and bedform dynamics	Insufficient reference support
Fast-water / riffle habitat	Habitat provision	Insufficient reference support
Sand + fines	Hyporheic connectivity, Sediment continuity	Insufficient reference support

Scores are computed automatically from the assessment's reference curves. Confirm state/region applicability of the curve source.